ECE/CSE 469 – Computer Design and Organization
HW #5

For each of these questions, assume we use 32-bit addresses.

1.) Consider the following sequence of byte addresses on a processor with 32 bit addresses:

24, 856, 700, 856, 24, 336, 260, 696, 256, 420, 340, 860

a.) For each of these references, identify the byte select, tag, and cache index given a direct-mapped cache with 16 4-byte blocks. Also, list if each reference is a hit or miss, assuming the cache is initially empty.

b.) For each of these references, identify the byte select, tag, and cache index given a direct-mapped cache with two-word blocks and a total size of eight blocks. Also, list if each reference is a hit or miss, assuming the cache is initially empty.

2.) You are given the following memory system:

Level	Hit Time	Hit Rate
L1	1 Cycle	90%
L2	5 Cycles	80%
Main Memory	100 Cycles	95%
Disk	100,000 Cycles	100%

For each of the following changes, indicate which level should be modified to yield the fastest system. Some changes may decrease performance over the original, but you must make the modification to one of the levels. The changes are separate (i.e. change b is on the original system, not on the one changed by change a). Note that the miss rate for the disk cannot be changed, so the right answer for c & d cannot be "disk".

a.)Decrease the hit time by a factor of 2.

b.) Increase the hit time by a factor of 2.

c.)Decrease the miss rate by a factor of 2.

d.) Increase the miss rate by a factor of 2.